

Probability and Random Processes

Random walks. Exercises

1. Two players, A and B , are engaged in a game. In each round, player A wins probability $p = 1/2$, receiving one unit from B , or loses with probability $q = 1/2$, paying one unit to B . Let X_n represent the net gain of A after the n -th round, where $X_0 = 0$.
 - (a) What is the probability of a draw in the n -th round?
 - (b) What is the expected number of draws in the first 6 games? [Hint: You may consider the indicator function I_k of a draw in the k -th game.]
 - (c) Provide an estimation of the probability that A has a net gain of at least 20 units in the 100-th round.
2. Two players, A and B , participate in a game where they toss a fair coin. If the result is heads, player A pays one unit to player B . Conversely, if the result is tails, player B pays one unit to player A . Let X_n represent the net gain of player A after the first n coin tosses.
 - (a) Compute $\mathbb{P}(X_3 = -1, X_5 = -3, X_9 = -3)$.
 - (b) If $X_{50} = 0$, what is the probability that $\mathbb{P}(X_{53} = -1, X_{55} = -3, X_{59} = -3)$.
 - (c) By using the Central Limit Theorem, give an approximate value for $\mathbb{P}(-196 \leq X_{10000} \leq 196)$.
3. Let $\{X_n : n \geq 0\}$ be a random walk with parameter $p = 3/4$.
 - (a) What is the probability of a return to the origin at time k ?
 - (b) If at time n we have $X_n = a$, what is the probability of visiting again a after k steps?
 - (c) Calculate the expected number of visits to the origin during the interval $1 \leq n \leq 6$. What is this number if $p = 1/4$?
4. Let $\{X_n : n \geq 0\}$ be a symmetric random walk. Calculate the following probabilities:
 - (a) That the random walk does not visit the coordinate -1 for any $1 \leq n \leq 2m$.
 - (b) That the random walk does not visit the coordinate -2 for any $1 \leq n \leq 2m$.
 - (c) That $X_n \leq 2$ for all $1 \leq n \leq 2m$,
5. A coin is tossed $n = 20$ times. Let Y be the random variable that counts the last time that the number of heads is equal to the number of tails ($Y = 0$ means that the game is not in a draw during the 20 tosses).
 - (a) Calculate $\mathbb{P}(Y = 2k)$.
 - (b) What is more probable, that the game has not been in a draw or that the last draw were for $k = 5$.
6. In an election, there are two candidates, A and B , and each of $2m$ voters votes with equal probability for A or B .
 - (a) If candidate A has received two votes more than candidate B , what is the probability that A has always been in the lead during the counting of the votes?
 - (b) What is the answer to the previous question if A has received $2r$ votes more than B ?
 - (c) What is the probability that one of the candidates has always been in the lead while the votes are being counted?

- (d) If candidate A has never fallen behind candidate B during the counting of the votes, calculate the probability that the counting finishes with an equal number of votes for both candidates.
7. A vehicle is in the plane at position $(0, 0)$. At every instant it moves with equal probability either to the North or to the East. We know that at instant $2n$ the vehicle is at position (n, n) . Find the probability that it has not passed through any point of the form (i, i) for $0 < i < n$.
8. At each time unit, a system is busy with probability $p = 1/2$ and idle with probability $q = 1 - p$. At time k , the performance level of the system, denoted as X_k , is defined as the difference between the number of busy times and the number of idle times.
- (a) What is the probability that the performance levels are 1, 2, and 3 at times 5, 8, and 15, respectively?
- (b) What is the probability that the performance remains positive for k consecutive time units if the performance starts at 1?
[Hint: You can use that if $\{Z_n: n \geq 0\}$ is a symmetric random walk, then $\mathbb{P}(Z_1 \neq 0, \dots, Z_{2m} \neq 0) = \mathbb{P}(Z_{2m} = 0)$.]
9. In a ballot, candidate A receives m votes and candidate B receives n votes, where $m > n$,
- (a) Prove that the number of voting sequences beginning with a vote for B is $\binom{m+n-1}{m}$.
- (b) Use a reflection argument to show that any voting sequence that starts with a vote for A and results in a tie at some point corresponds one-to-one with a voting sequence that starts with a vote for B .
- (c) From (a) and (b), deduce that the probability that candidate A was always ahead of candidate B during the vote counting is $(m - n)/(m + n)$.
10. Let $\{X_n: n \geq 0\}$ be a symmetric random walk such that $X_0 = 0$. Prove that if a is a positive integer, then

$$\mathbb{P}(X_1 \neq 0, \dots, X_{n-1} \neq 0, X_n = a) = \frac{a}{n} \mathbb{P}(X_n = a).$$

[Hint: consider the trajectories from $(1, 1)$ to (n, a) in the (n, X_n) -plane representation of the random walk.]

Answers

1. (a) $\binom{2k}{k} \left(\frac{1}{2}\right)^{2k}$ (b) $19/16$ (c) 0.0228
2. (a) $9/256$ (b) $9/256$ (c) 0.950
3. If k is even, $\left(\frac{k}{2}\right) p^{k/2} q^{k/2}$ (b) the same (c) $735/1024$ if $p = 3/4$; if $p = 1/4$, the same.
4. (a) $\binom{2m}{m} 2^{-2m}$ (b) $\binom{2m+1}{m} 2^{-2m}$ (c) $\frac{3m+1}{m+1} \binom{2m}{m} 2^{-2m}$
5. (a) $\binom{2k}{k} \binom{20-2k}{10-k} / 2^{20}$ (b) That the game has not been in a draw
6. (a) $1/m$ (b) r/m (c) $\binom{2m}{m} \frac{1}{2^{2m}}$ (d) $1/(m+1)$
7. $1/(2n-1)$
8. $525/16384$ (b) $\binom{2m}{m} 2^{-2m}$ if $k = 2m$, $\binom{2m+2}{m+1} 2^{-2m-2}$ if $k = 2m+1$
- 9.
- 10.